2D-PRINTER

Intern Project For Group-5

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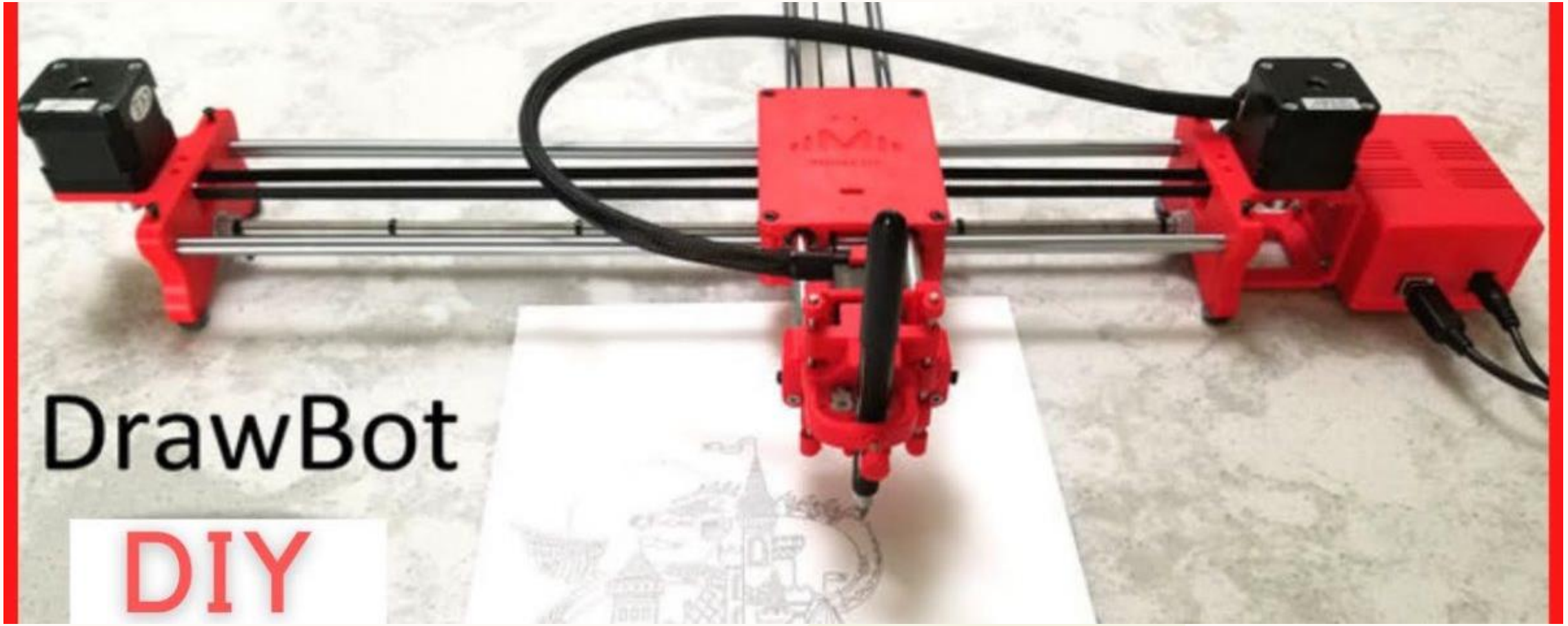
INTRODUCTION:

This Drawing Robot/Pen Plotter/Drawing Machine is similar to the commercially available AxiDraw. It is powered by an Esp 32 controller, uses a CNC Shield, and GRBL firmware.

The cost to build the Drawing Robot is between 5000 rupees to 7000 rupees depending on where you buy your parts.

THE MODEL WE WILL BE FOLLOWING:



THE FLOW OF WORK

- *Firstly image will be converted into g-code by using ink space and then the g-code will be edited for the microcontroller.*
- *The microcontroller will thereby instruct the CNC shield and the motor drivers for the movement of motors which will move the pen accurately according to the coordinates fetched by the g-codes.*
- *The stepper motors will be responsible for the lifting and dropping of the pen.*
- *Servo motors will be responsible for the pen in different axis.*

PARTS AND MATERIALS REQUIRED:

- *2 x Nema 17 Stepper Motors*
- *2 x Linear Rod M8 x 450mm for X Axis*
- *2 x Linear Rod M8 x 350mm for Y Axis*
- *2 x Linear Rod 3mm for Z Axis*
- *1 x Threaded Rod M8 x 480mm*
- *8 x LM8UU Bearings*
- *1 x Servo Sg90*
- *1 x Spring 5m (from ball point pen)*
- *2 x GT2 Pulley, 16 teeth*
- *5 x Bearing 624zz*
- *1 x 2000mm GT2 belt*
- *1 x Arduino Uno*
- *1 x CNC Shield*
- *2 x A4988 Stepper driver with heatsink*
- *6 x Jumpers*
- *1 x 12V 2A Power Supply*

PARTS AND MATERIALS REQUIRED: (CONTINUATION)

Nuts

- 7 x M3-0.5
- 5 x M4-0.7
- 4 x 5/16in-18

Screws

- 13 x Phillips M3-0.5 x 16mm
- 4 x Phillips M3-0.5 x 6mm
- 5 x Phillips M4-0.7x 35mm
- 1 x Hex M3-0.5 x 20mm

Washers

- 4 x 5/16in washer
- 4 x M3 washers

WORK DISTRIBUTION:

Mechanical section :

Handled by me, since 3D printing is accessible. Firstly the mechanical parts are being bought and then 3D printing will be performed. Assembly will also be done if possible, else we'll do it together in college.

Coding Section :

Handled by both Hameed and Shantanu, as instructed by our seniors. They will be studying the basics of OpenCV including image processing and then convert the image into G-code which will guide our pen. It'll specify some starting and ending coordinates for the pen to move.

PROJECT CONCLUSION:

A 2D printer robotic project typically involves creating a machine that can interpret computer-generated designs and replicate them onto a physical surface. Such a project might include assembling a robotic plotter that operates on two axes, X and Y, to draw images or patterns. The core of the system could be an embedded platform like Esp 32, which controls stepper motors to move the plotter's pen across the drawing surface. The pen's movement would be based on Computer Numerical Control (CNC) principles, allowing for precise and automated drawing tasks. This type of project is not only fascinating for hobbyists but also serves educational purposes, providing hands-on experience with robotics, programming, and mechanical design.

THANK YOU

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